

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

II Semester of MSc Biotechnology/ Microbiology/ Biochemistry Examination, April 2019

Course code & Name: MS762 Immunology

29.04.2019

Monday

10:00 AM To 01:00 PM

Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

20

1. Chickens are naturally immune against anthrax, due to
 - (i) very low gastric pH
 - (ii) high body temperature
 - (iii) large number of feathers on the body
 - (iv) antibodies passed by mother to egg
2. GALTs are found in
 - (i) Bones
 - (ii) Intestine
 - (iii) Liver
 - (iv) Spleen
3. An "allotype" reveals
 - (i) CDR of an Ab
 - (ii) Multiple alleles of an Ab
 - (iii) Constant region determinants of Ab
 - (iv) None of the above
4. MHC stands for
 - (i) Major Histocomparability Complex
 - (ii) Minor Histocomparability Complex
 - (iii) Major Histocompatibility Complex
 - (iv) Minor Histocompatibility Complex
5. Select the right combination
 - (i) ADCC/ B-cells
 - (ii) ADCC/ NK cells
 - (iii) ADCC/ RBCs
 - (iv) All the above are right

6. An individual with homozygous recessive “hh” genotype represent
 - (i) Presence of both Bombay and ABO blood group
 - (ii) Presence of Bombay and absence ABO blood group
 - (iii) Absence of Bombay and presence of ABO blood group
 - (iv) Absence of both Bombay and ABO blood group
7. B-cell immunodeficiency is a type of
 - (i) Humoral immunodeficiency
 - (ii) Cell-mediated immunodeficiency
 - (iii) Both (i) and (ii)
 - (iv) Neither (i) nor (ii)
8. Antibodies are secreted by
 - (i) T-cells
 - (ii) Monocytes
 - (iii) Plasma cells
 - (iv) RBCs
9. Select the right order of development
 - (i) Hematopoietic stem cell>Monocyte>Myeloid progenitor>Myeloblast
 - (ii) Hematopoietic stem cell>Myeloblast>Myeloid progenitor>Monocyte
 - (iii) Hematopoietic stem cell>Myeloid progenitor> Monocyte>Myeloblast
 - (iv) Hematopoietic stem cell>Myeloid progenitor>Myeloblast>Monocyte
10. Complement molecules are _____ in nature
 - (i) Lipoprotein
 - (ii) Nucleoprotein
 - (iii) Protein
 - (iv) Glycoprotein
11. ABO blood group antigens are
 - (i) Proteins
 - (ii) Lipids
 - (iii) Glycoproteins
 - (iv) Glycolipids
12. Select the most appropriate combination series
 - (i) TCR-(Ag)-MHC
 - (ii) MHC-(TCR)-Ag
 - (iii) Ag-(MHC)-TCR
 - (iv) TCR-(MHC)-Ag
13. DTH is type _____ hypersensitivity
 - (i) I
 - (ii) II
 - (iii) III
 - (iv) IV
14. SCID can be associated with
 - (i) Transplantation immunology
 - (ii) Autoimmunity
 - (iii) Immunodeficiency
 - (iv) Hypersensitivity
15. Against which cancer, vaccine is available?
 - (i) Cervical cancer
 - (ii) Liver cancer
 - (iii) Leukaemia
 - (iv) None of the above
16. Which of the following is not an immunodiagnostic technique
 - (i) FACS
 - (ii) AlphaLISA
 - (iii) RIA
 - (iv) None of the above
17. Fusion of myeloma cell with T-lymphocyte results in mAbs production

True False
18. Both membrane-bound and secretory antibodies are glycoprotein in nature

True False
19. Primary immune response slower in infants and faster in adults as compared to secondary immune response

True False
20. Due to inflammation there is long-term loss of function of the affected part

True False

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Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Answer the following questions as directed **20**

1. Attempt **ANY THREE** **15**

- (i) Which type of molecules bring antigenic peptide near to a specific type of immune cell. Name the molecules and the specific type of cell. Compare the structure of the said molecules.
- (ii) What is HAMA and how it can be avoided? Describe different types of genetically engineered Abs.
- (iii) What is tolerance in immunology? Discuss the mechanisms that may induce autoimmunity in an individual.
- (iv) Explain various phases of cancer immunoediting.

2. (i) Explain, how mAbs are produced? **3**

OR

(ii) Discuss MALT

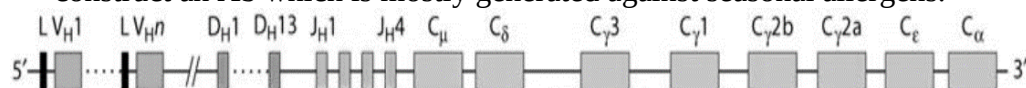
3. An individual was diagnosed with hepatitis, due to HBV infection. 2
The infection persisted because the T-cells of the host could not destroy the infected hepatocytes due to non-presentation of viral peptides. Name any four molecules and/ or molecular complex involved in viral peptide presentation that could have gone abnormal.

SECTION – II

Q-III Answer the following questions as directed **30**

1. Attempt **ANY THREE** **15**

- (i) Systematically rearrange the various gene segments of your choice and construct an Ab which is mostly generated against seasonal allergens.



- (ii) Describe inflammation with an especial focus on the role of neutrophils.
- (iii) Compare the complement pathways that have dissimilar source of activation but similar mechanism of action.
- (iv) Compare (a) innate and adaptive immunity; (b) humoral and cell mediated immunity.

2. Attempt **ANY THREE** **15**

- (i) What do you understand by vaccination? Describe sub-unit vaccines.
- (ii) Describe the role of ABO and H alleles in ABO and Bombay blood grouping.
- (iii) What is erythroblastosis fetalis? Describe, how it can be prevented?
- (iv) Describe generation and activation of BCR containing cells and the role played by BCR.